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CASE STUDY

PetroFlow — Enterprise Oil & Gas Pipeline PMO & Industrial Asset Management Platform

Unified PMO governance, specialist workload heatmaps, P&ID drawing version control, AI-assisted proposal reporting, and SCADA pipeline telemetry — integrated into a single digital system of record.

CLIENT	National Energy & Infrastructure Operations Enterprise
INDUSTRY	Oil & Gas Infrastructure, Pipeline Engineering & Industrial Asset Management
SOLUTION	PetroFlow Industrial PMO, Asset Telemetry & Compliance Suite
PREPARED BY	Vishal N Team · Cyber Infrastructure Pvt. Ltd.

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SERVICE CATEGORIES UNIFIED

100%

DIGITAL APPROVAL TRAIL

AI

COPILOT & AUTOMATION

LIVE

SPECIALIST WORKFORCE
HEATMAP

1. Executive Summary

PetroFlow delivers end-to-end PMO and industrial asset operations for major energy infrastructure projects including regional refinery HVAC refits, subsea power grid integrations, and high-pressure gas transmission pipelines. Before this engagement, operations ran on fragmented Excel workbooks, disconnected email approval trails, and unmonitored safety risk logs — leaving executive leadership without real-time visibility into project health, specialist workload allocations, or regulatory HSE compliance.

Cyber Infrastructure Pvt. Ltd. engineered **PetroFlow** — a unified, full-stack industrial engineering PMO platform that integrates executive BI dashboards, specialist capacity heatmaps, P&ID drawing version control (v1, v2, v3), AI-assisted proposal reporting, and a SCADA pipeline telemetry extension into one central system of record.

9

DISCIPLINES UNIFIED

100%

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RESOURCE ALLOCATION

What the Platform Replaces

● Manual Excel Project Tracking

Replaced three daily disconnected status spreadsheets with a centralized Executive BI Portfolio dashboard (/dashboard) tracking SAR/INR budgets, milestones, and project status.

● Paper-Based Approvals

Transformed manual sign-off delays into a multi-stage Digital Approval Workflow engine (/approvals) with time-stamped audit logs and instant status transitions.

● Scattered Drawing Versioning

Unified P&ID drawings, CAD models, and BIM specs into a version-controlled DMS (/documents) supporting v1, v2, v3 drawing revisions with instant browser preview.

● Informal Risk Tracking

Replaced unmonitored risk logs with a scored Safety Risk Register (/risks) evaluating Severity × Likelihood alongside HSE compliance audit checklists.

2. Client Profile & Legacy Pain Points

PetroFlow manages oil and gas engineering projects across **Gas Transmission Pipelines, Refinery HVAC Systems, Subsea Grid Integration, P&ID Schematics, and Safety Compliance**. Operational volume expansion across multi-regional field sites introduced critical bottlenecks:

- **Refinery & Pipeline Blind Spots:** Lack of real-time visibility forced executive management to evaluate facility performance only after shift ends or milestone deadlines passed.
- **Specialist Overallocation:** Piping and mechanical engineers were assigned without workload capacity tracking, leading to undetected burnout (>100% capacity) and scheduling delays.
- **P&ID Version Conflicts:** Technical drawings and schematics were shared via email, leading to conflicting drawing revisions on field construction sites.
- **Unmonitored Safety Risks:** Environmental compliance, gas pressure breaches, and budget overruns were tracked manually, preventing proactive mitigation before critical audits.

WHY IT MATTERED

Without a unified digital intelligence layer, each engineering discipline operated as an isolated island — preventing real-time portfolio comparison, resource leveling, and cross-project HSE compliance auditing.

3. Solution Architecture

Cyber Infrastructure Pvt. Ltd. engineered PetroFlow as a modular, full-stack PMO and manufacturing intelligence platform, purpose-built to unify multi-discipline engineering operations into a single source of operational truth.

MODULE	CAPABILITIES	ENDPOINT
Executive Portfolio Dashboard	Consolidated 9-discipline project health index; live budget vs. spent tracking; AI executive summaries; 90-day progress trends.	/dashboard
Pipeline PMO & Projects	Multi-phase deliverable tracking (Design, Approval, Execution, Handover); milestone schedules; role-gated task priority controls.	/projects
Specialist Heatmap & Crew	Workload allocation % (0-100%+); overallocation alert badges; discipline filtering & Field Staff Directory.	/workforce
Pipeline DMS Drawings	Version-controlled P&ID drawing registry (v1, v2, v3); discipline tagging; OCR smart search; in-browser PDF/drawing preview.	/documents
Safety Risks & Compliance	Scored risk matrix (Severity × Likelihood); mitigation tracking; HSE regulatory compliance checklists.	/risks
Workflow Automation	Multi-stage digital sign-off workflows; step-by-step approval trails; automated status updates.	/approvals
Client CRM & Proposals	Bidding pipeline & lead tracking; contract value management; AI-assisted proposal generator.	/proposals
Pipeline AI Copilot	Natural language database query engine; automated project health summaries; risk mitigation advisor.	/ai-assistant
Pipeline IoT & Telemetry	Live SCADA pressure/flow telemetry map; segment degradation grid; predictive maintenance alerts & incident control desk.	/pipeline/map

4. Core Module Deep Dive

A. Executive Control Tower & Live Project Health

Executive leadership receives a side-by-side comparison of active engineering projects across regional refinery refits, subsea grids, and transmission pipelines. Project health is calculated in real time:

Project Health = Budget Efficiency × Schedule Compliance × Risk Index

- B. Safety Risk Matrix & HSE Compliance**

Continuous risk score evaluation against project milestones. When high-severity risks emerge, PetroFlow automatically links regulatory HSE compliance checklists and alerts designated PMO Directors.

C. PetroFlow AI Copilot & Natural Queries

Natural language AI engine processes queries against live database tables (e.g., "Which projects are At-Risk?") and automatically generates bid proposals and executive summaries.
- D. SCADA Telemetry & Predictive Maintenance**

Provides real-time IoT and SCADA monitoring for pipeline assets. Tracks pressure (bar/PSI) and flow rates, ranks high-wear pipeline segments, and dispatches automated work orders to field crews before failure occurs.

5. Technology Stack & Role Scoping

LAYER	TECHNOLOGIES & TOOLS
Frontend Framework	React 18, React Router v6, Vite
Styling & Design System	Vanilla CSS, TailwindCSS, Lucide Icons, Glassmorphic UI & Dark Mode
Data Visualization	Recharts (Financial trends, risk distribution, specialist utilization heatmaps)
Backend API Engine	Node.js, Express.js (RESTful Architecture, JWT Auth, Multer Disk Storage)
Database & ORM	SQLite / MySQL with Sequelize ORM — 12 Relational Data Models
AI Integration	Natural Language Query Engine + LLM API Connectors (OpenAI / Gemini SDK)
Document Processing	Multer Disk Storage, File System Static Streaming, Smart OCR Search
Containerization & Infra	Docker & Docker Compose orchestrating Express Backend, React Client, Database

Role-Based Access Control Scoping

PMO Director Full portfolio oversight, budget approvals, and global task priority management	Project Manager Project creation, deliverable scheduling, risk mitigation, and team allocation	Specialist Engineer Task execution, P&ID revision uploads, and status logging
Field Inspector Telemetry monitoring, inspection audit logging, and incident reports	Finance Lead Commercial proposal approval, contract value tracking, and budget auditing	System Admin Role management, system configuration, user provisioning, and audit logs

6. Metrics & Conclusion

- **Multi-Discipline Engineering & Infrastructure Support:** Unified coverage across Refinery Refit, Subsea Grid Integration, and Gas Transmission lines.
- **100% Digital Approval & Audit Trail:** Replaced manual paper approvals with time-stamped digital workflows and version-controlled DMS logs.
- **Live SCADA Telemetry & Predictive Maintenance:** Continuous sensor streaming for pipeline pressure, flow rates, and segment degradation alerts.
- **Role-Based Access Across 6 User Types:** Scoped permissions from PMO Directors down to Field Inspectors and Technical Specialists.

7. CONCLUSION

The PetroFlow platform replaces a patchwork of spreadsheets and manual paper approval trails with a single connected digital operations layer. Built on a modern full-stack architecture with AI integration at its core, PetroFlow gives engineering consultancies and oil & gas operators a scalable foundation to manage complex project portfolios, enforce HSE compliance, and optimize workforce allocation.

DELIVERED BY: Vishal N Team · Cyber Infrastructure Pvt. Ltd.